MSI





CAPS

REVISION

DIFFERENTIATION

GRADE 12

CONCEPT	HOW TO LEARN IT?	RELEVANT FORMULAE AND KEYWORDS
Finding the average gradient	Practice by substituting into the formula.	$m_{avr} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$
	This is only working out the average gradient between two points. The y-values are not given but will have to be calculated first before substituting into the formula.	Average gradient, gradient between two points, gradient of $f(x)$ between two givens x values.
	Practice by doing sums.	Explain the concept in own words and doing the sum.
	Understands that average gradient differs from first principles in such a way that there is a distance between the two x values. Make sure you understand average gradient.	Can you recognise this concept in a paper?
	This is linked to Calculus application question where you will be asked to find the average rate of change of for example the growth of bacteria cells.	
Finding the derivative by using first principles	Practice various examples of the following functions. $f(x) = ax^2$ $f(x) = \frac{k}{x}$	$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ Formula for gradient at one point, $(x; f(x))$ Also called the derivative.

	$f(x) = ax^3$ Substitute into the formula. Be very careful with the use of brackets and the correct use of the distributive property of multiplication. When two points of the average gradient get closer and closer to each other (converge), then the limit of the average gradient exists.	Also called finding the derivative using "the definition".
	How to substitute into the correct formula. The correct $f'(x)$ notation. How to use the distributive property of multiplication.	Derivative using first principles, ground principles, finding the derivative using the definition
	Start with doing the classwork. Then do homework sums. Practice plenty examples on each type.	You can check your answer by quickly doing the sum using the rules of differentiation and if it's the same you know you are doing it right. No level 7 type questions
	This question is freestanding. Must just be able to calculate the answer.	
Rules of differentiation	Make sure you understand the rule $x^n = nx^{n-1}$ You may not differentiate when there are brackets – first multiply it out. You may not differentiate when x is the denominator – write it as x^{-1}.	Different notations: $\frac{d_y}{d_x}$ if $y =$ ... is given $D_x[\ldots\ \ldots]$ $f'(x)$ if $f(x)$ is given

	You may not differentiate when there is a surd- write it as $\sqrt{x} = x^{\frac{1}{2}}$. Always leave your answers with positive exponents Study this by practicing all the sums you can find. The more you do, the better you will become.	Rules of differentiation: $\frac{d_y}{d_x}[f(x) + g(x)] = \frac{d_y}{d_x}f(x) + \frac{d_y}{d_x}g(x)$ $\frac{d_y}{d_x} = [f(x) - g(x)] = \frac{d_y}{d_x}f(x) - \frac{d_y}{d_x}g(x)$ $\frac{d_y}{d_x}kf(x) = k\frac{d_y}{d_x}f(x)$; k is a constant $\frac{d_y}{d_x}k = 0$; k is a constant
	How to apply the rules for differentiation? The special cases that will be given to me, where I need to work with the equation first to turn it into an equation where I can differentiate with respect to x.	Differentiate. Differentiate with respect to x, Find the derivative using rules, $D_x[\ldots\ \ldots]$
	The more sums you practice, the better you will become!	If you know the rule and are able to apply it correctly, you know you understand.
	Look for intricate fractions and sums that contain surds. Practice a variety of different sums.	If the sum looks intricate with difficult fractions and surds, and you can get the answer, you know you have mastered this section.

	You can use the rules of differentiation anywhere in the syllabus when you are required to find the gradient of a tangent to a curve, or the x- co-ordinate of a local maximum or minimum stationary / turning point of a function.	
Finding the equation of a tangent to a curve or function	You have to make sure you can apply the rules of differentiation first. Differentiate the function to find the expression for the gradient, using the rules of differentiation. Substitute the x-value of the point of contact into the gradient expression to obtain the value of the gradient. Substitute the x-value into the original formula to get the corresponding y-value at the point of contact. Use the gradient and the point of contact to determine the equation of the tangent. Then you practice this by doing plenty of sums.	$y = mx + c$ $y - y_2 = m(x - x_1)$
	First you must understand why you are doing this question in this topic. When you find the derivative, the answer you get means the "gradient of the tangent to the curve, at any point." First find the gradient by using the rules of differentiation.	Equation of tangent to function.

	Then calculate the corresponding y- coordinate at the given x. Then substitute into the formula of a straight line to get the equation.	
	Make sure you are able to work out the equation of a normal straight line. Once you can master this, it is just a matter of understanding that you use the rules of differentiation to find the gradient now, you don't need the gradient formula here.	If you are able to find the correct equation of the tangent, you understand.
	Understand the concept of finding the derivative means "the gradient of the tangent to a curve" at any given point on the cure. Make sure you can use the rules of differentiation to find the gradient of the tangent, and that you are able to work out the equation of this line accurately.	You must be able to recognise this question in a paper. Don't think it will be asked straight forward. Sometimes they swop it around, by giving you the gradient, and then you have to calculate at which x-co-ordinate the tangent is. Then find the equation.
	This is a skill you must master – finding the equation of a tangent. This can also appear in Paper 2 in the Analytical Geometry section.	
The cubic function	First you have to sketch about ten of these functions to really get a feel of how this works. You must be able to plot giving attention to x-intercepts, y-intercepts and stationary points (turning points) by using rules of differentiation to get the x-value of the turning point. After sketching you can find the equation of the cubic function.	Three different roots are given. $$y = a(x - x_1)(x - x_2)(x - x_3)$$ Two roots, and one is also the turning point. $$y = a(x - x_1)(x - x_2)^2$$

	There are 5 different scenarios. Make sure you are familiar with these. Once you can sketch and find the equation you can move on to deductions of graphs. You must know all the previous theory of graphs such as domain, range etc.	
	How to sketch a cubic function? The shape is influenced by the sign of the "a" value. How to work out the turning points by using rules of differentiation. How to work out the points of inflection using the second derivative. The theory on concavity. How to find the equation of a cubic function.	Cubic function Stationary points Point of inflection Concavity The derivative over a specific interval. Local maximum and minimum. x-intercepts, roots y-intercept
	You sketch different cubic functions first, then you calculate the equations and then you do the questions on deductions.	If you can sketch a cubic function, find the equation and do the deductions then you understand.
	Do plenty of different examples at home independently.	You will only know this if you do sums from previous examination papers.

	Know the basics very well and familiarise yourself with a variety of different examples.	
	This topic can be integrated with finding the equation of a tangent to the function. You must also be able to relate the knowledge of a cubic function's maximum and minimum value to Calculus applications. For example, if the rate at which water is flowing into a dam is given by $f(x) = x^3 + x^2 - 4x + 7$, then the maximum time it will take to fill up, will be calculated by finding the derivative. A rough sketch of the cubic function will indicate at which x the maximum will be.	
Calculus application	You need to practice all the different sums you can find from previous examination papers. You will apply all your knowledge from previous grades, especially formulas for perimeter, area etc.	Perimeter of Rectangle $= 2(l + b)$ Area of rectangle $= l \times b$ Perimeter of square $= 4l$ Area of square $= l^2$ Circumference of circle $= 2\pi r$ Area of circle $= \pi r^2$ Area of a right- angled triangle $= \frac{1}{2} \cdot b \cdot h$
	You will apply all your knowledge from previous grades, especially formulas for perimeter, area etc. You need to understand all the theory of Calculus and finding the maximum or minimum values. You need to understand rate of change.	Maximum and minimum Speed Acceleration Rate of change

	When the height is given, the first derivative will give you speed and the second derivative will give you acceleration.	
	You have to realise that this is not the section where you study the night before!! You have to do plenty of examples and different sums in order for you to master this section. You must be aware of the fact that this question will be the unseen question. It does not matter how many you have practiced the one in the examination will be different to all the ones you have practiced. The key to success here is that if you fully understand all the concepts of Calculus and you know your formulas, then you will be able to answer the question.	If you are able to explain the Calculus theory and apply your knowledge you will do well n this question.
	Practice, practice, practice!!! Expect to use ALL your knowledge from previous grades and apply it here.	Work through as many previous examination papers as you possibly can.
	This is the "where do you find Maths in real life" question. Expect a different sum to what you have seen before. Know your knowledge of Calculus and the theory on maximum and minimum. Know that if you derive height, you get instantaneous speed. If you derive speed you get acceleration. Know all your formulas on area and perimeter and be able to apply it here on maximum and minimum.	

FIRST PRINCIPLE

Activity 1

1.1 Determine $f'(x)$ from first principles if $f(x) = -3x^2$. (5)

1.2 Differentiate f by first principles where $f(x) = x^2 - 2x$. (5)

1.3 Differentiate f from first principles if it is given that $f(x) = -5x^2 + 2$. (5)

1.4 Differentiate$f(x)$ from first principles if $f(x) = -2x^2 + 3$. (5)

1.5 Differentiate f from first principles: $f(x) = \frac{1}{x}$ (4)

1.6 Differentiate $g(x) = x^2 - 5$ from first principles. (5)

Activity 2

2.1 Use the definition of differentiate $f(x) = 1 - 3x^2$. (Use first principles.) (4)

2.2 Determine $f'(x)$ from first principles if $f(x) = -4x^2$. (5)

2.3 Determine $f'(x)$ from first principles if $f(x) = 9 - x^2$. (5)

2.4 Determine $f'(x)$ from first principles if $f(x) = 2x^2 - 5$. (5)

2.5 If $f(x) = -x^2$, determine $f'(x)$, by using first principles. (5)

Activity 3

3.1 Determine the derivative of $f(x) = 2x^2 + 4$ from first principles. (4)

3.2 Use the definition of the derivative (first principles) to determine $f'(x)$ if $f(x) = 2x^3$. (5)

3.3 If $f(x) = x^2 - 3x$, determine $f'(x)$ from first principles. (5)

3.4 Given: $f(x) = -\frac{2}{x}$

3.1 Determine $f'(x)$ from first principles. (5)

3.2 For which value(s) of x will $f'(x) > 0$? Justify your answer. (2)

3.5 If $f(x) = -5x^2$, determine $f'(x)$ from first principles. (5)

RULES OF DIFFERENTIATION

Activity 4

4.1 Determine, using the rule of differentiation: $\frac{dy}{dx}$ if $y = \frac{\sqrt{x}}{2} - \frac{1}{6x^3}$. Show ALL Calculations. (3)

4.2 Evaluate $D_x[(x^3 - 3)^2]$ (3)

4.3 Determine $\frac{dy}{dx}$ if $y = \frac{4}{\sqrt{x}} - \frac{x^3}{9}$ (3)

4.4 Use the rules of differentiation to determine the following: $D_x[(x - 2)(x + 3)]$ (3)

4.5 Evaluate: $\frac{dy}{dx}$ if $y = x^2 - \frac{1}{2x^3}$ (2)

Activity 5

5.1 Use the rules of differentiation to determine $\frac{dy}{dx}$ if $y = (2 - 5x)^2$ (3)

5.2 Evaluate $\frac{dy}{dx}$ if $y = \frac{x^6}{2} + 4\sqrt{x}$. (3)

5.3 Calculate $D_x\left[4 - \frac{4}{x^3} - \frac{1}{x^4}\right]$. (3)

5.4 Determine $\frac{dy}{dx}$ if $y = \left(1 - \sqrt{x}\right)^2$. (3)

5.5 Evaluate $\frac{dy}{dx}$ if $y = \frac{3}{2x} - \frac{x^2}{2}$. (3)

5.6 Determine $f'(1)$ if $f(x) = (7x + 1)^2$ (4)

Activity 6

6.1 Evaluate $D_x\left[1 + 6\sqrt{x}\right]$ (2)

6.2 Determine $\frac{dy}{dx}$ if $y = \frac{8 - 3x^6}{8x^5}$ (4)

6.3 Evaluate $\frac{dy}{dx}$ if $y = x^{-4} + 2x^3 - \frac{x}{5}$. (3)

6.4 Determine $D_x[a^2x^2 + x]$. (2)

6.5 Determine $f'(x)$ if: $f(x) = \frac{x^2 - 9}{x - 3}$ (4)

6.6 Evaluate $f(x) = \sqrt{x}\left(x + \frac{1}{x}\right)^2$. (4)

DIFFERENTIATION FROM FIRST PRINCIPLES

Activity 1

1.1	$f(x) = -3x^2$ $f(x+h) = -3(x+h)^2$ $f(x+h) = -3x^2 - 6xh - 3h^2$ $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{-3x^2 - 6xh - 3h^2 - (-3x^2)}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{-3x^2} - 6xh - 3h^2 + \cancel{3x^2}}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{h}(-6x - 3h)}{\cancel{h}}$ $f'(x) = -6x - 3(0)$ $f'(x) = -6x$
1.2	$f(x) = x^2 - 2x$ $f(x+h) = (x+h)^2 - 2(x+h)$ $f(x+h) = x^2 + 2xh + h^2 - 2x - 2h$ $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{x^2 + 2xh + h^2 - 2x - 2h - (x^2 - 2x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{x^2} + 2xh + h^2 \cancel{- 2x} - 2h \cancel{- x^2} + \cancel{2x}}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{h}(2x + h - 2)}{\cancel{h}}$ $f'(x) = 2x + 0 - 2$ $f'(x) = 2x - 2$

1.3	$f(x) = -5x^2 + 2$ $f(x+h) = -5(x+h)^2 + 2$ $f(x+h) = -5x^2 - 10xh - 5h^2 + 2$ $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{-5x^2 - 10xh - 5h^2 + 2 - (-5x^2 + 2)}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{-5x^2} - 10xh - 5h^2 \cancel{+ 2} + \cancel{5x^2 - 2}}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{h}(-10x - 5h)}{\cancel{h}}$ $f'(x) = -10x - 5(0)$ $f'(x) = -10x$
1.4	$f(x) = -2x^2 + 3$ $f(x+h) = -2(x+h)^2 + 3$ $f(x+h) = -2x^2 - 4xh - 2h^2 + 3$ $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{-2x^2 - 4xh - 2h^2 + 3 - (-2x^2 + 3)}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{-2x^2} - 4xh - 2h^2 \cancel{+ 3 + 2x^2 - 3}}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{h}(-4x - 2h)}{\cancel{h}}$ $f'(x) = -4x - 2(0)$ $f'(x) = -4x$
1.5	$f(x) = \frac{1}{x}$ $f(x) = \frac{1}{(x+h)}$ $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$

	$f'(x) = \lim_{h\to 0}\left[\frac{1}{(x+h)} - \frac{1}{x}\right] \div h$ $f'(x) = \lim_{h\to 0}\left[\frac{x-(x+h)}{x(x+h)}\right] \div h$ $f'(x) = \lim_{h\to 0}\left[\frac{\cancel{-h}}{x(x+h)}\right] \times \frac{1}{\cancel{h}}$ $f'(x) = \lim_{h\to 0}\left[\frac{-1}{x(x+h)}\right]$ $f'(x) = \frac{-1}{x(x+0)}$ $f(x) = -\frac{1}{x^2}$
1.6	$f(x) = x^2 - 5$ $f(x+h) = (x+h)^2 - 5$ $f(x+h) = x^2 - 2xh - h^2 - 5$ $f'(x) = \lim_{h\to 0}\frac{f(x+h) - f(x)}{h}$ $f'(x) = \lim_{h\to 0}\frac{x^2 - 2xh - h^2 - 5 - (x^2 - 5)}{h}$ $f'(x) = \lim_{h\to 0}\frac{\cancel{x^2} - 2xh - h^2 \cancel{- 5 - x^2 + 5}}{h}$ $f'(x) = \lim_{h\to 0}\frac{\cancel{h}(-2x-h)}{\cancel{h}}$ $f'(x) = -2x - (0)$ $f'(x) = -2x$

Activity 2

2.1	$f(x) = 1 - 3x^2$ $f(x+h) = 1 - 3(x+h)^2$ $f(x+h) = 1 - 3x^2 - 6xh - 3h^2$ $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{1 - 3x^2 - 6xh - 3h^2 - (1 - 3x^2)}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{1 - 3x^2} - 6xh - 3h^2 \cancel{- 1 + 3x^2}}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{h}(-6x - 3h)}{\cancel{h}}$ $f'(x) = -6x - 3(0)$ $f'(x) = -6x$
2.2	$f(x) = -4x^2$ $f(x+h) = -4(x+h)^2$ $f(x+h) = -4x^2 - 8xh - 4h^2$ $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{= -4x^2 - 8xh - 4h^2 - (-4x^2)}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{-4x^2} - 8xh - 4h^2 \cancel{+ 4x^2}}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{h}(-8x - 4h)}{\cancel{h}}$ $f'(x) = -8x - 4(0)$ $f'(x) = -8x$
2.3	$f(x) = 9 - x^2$ $f(x+h) = 9 - (x+h)^2$ $f(x+h) = 9 - x^2 - 2xh - h^2$

	$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{9-x^2-2xh-h^2-(9-x^2)}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{9-x^2}-2xh-h^2\cancel{-9+x^2}}{h}$ $f'(x) = \lim_{h\to 0} \frac{h(-2x-h)}{h}$ $f'(x) = -2x-(0)$ $f'(x) = -2x$
2.4	$f(x) = 2x^2-5$ $f(x+h) = 2(x+h)^2-5$ $f(x+h) = 2x^2+4xh+2h^2-5$ $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{2x^2+4xh+2h^2-5-(2x^2-5)}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{2x^2}+4xh+2h^2\cancel{-5-2x^2+5}}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{h}(4x+2h)}{\cancel{h}}$ $f'(x) = 4x+2(0)$ $f'(x) = 4x$
2.5	$f(x) = -x^2$ $f(x+h) = -(x+h)^2$ $f(x+h) = -x^2-2xh-h^2$ $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{-x^2-2xh-h^2-(-x^2)}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{-x^2}-2xh-h^2\cancel{+x^2}}{h}$

	$f'(x) = \lim_{h\to 0} \frac{h(-2x-h)}{h}$ $f'(x) = -2x - (0)$ $f'(x) = -2x$

Activity 3

3.1	$f(x) = -2x^2 + 4$ $f(x+h) = -2(x+h)^2 + 4$ $f(x+h) = -2x^2 - 4xh - 2h^2 + 4$ $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{-2x^2 - 4xh - 2h^2 + 4 - (-2x^2 + 4)}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{-2x^2} - 4xh - 2h^2 + 4 \cancel{+ 2x^2 - 4}}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{h}(-4x - 2h)}{\cancel{h}}$ $f'(x) = -4x - 2(0)$ $f'(x) = -4x$
3.2	$f(x) = 2x^3$ $f(x+h) = 2(x+h)^3$ $f(x+h) = 2(x+h)(x+h)^2$ $f(x+h) = 2x^3 + 6x^2h + 6xh^2 + 2h^3$ $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{2x^3 + 6x^2h + 6xh^2 + 2h^3 - (2x^3)}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{2x^3} + 6x^2h + 6xh^2 + 2h^3 \cancel{- 2x^3}}{h}$

	$f'(x) = \lim_{h\to 0} \frac{\cancel{h}(6x^2 + 6xh + 2h^2)}{\cancel{h}}$ $f'(x) = 6x^2 + 6x(0) + 2(0)^2$ $f'(x) = 6x^2$
3.3	$f(x) = x^2 - 3x$ $f(x+h) = (x+h)^2 - 3(x+h)$ $f(x+h) = x^2 + 2xh + h^2 - 3x - 3h$ $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{x^2 + 2xh + h^2 - 3x - 3h - (x^2 - 3x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{x^2} + 2xh + h^2 \cancel{- 3x} - 3h \cancel{- x^2 + 3x}}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{h}(2x + h - 3)}{\cancel{h}}$ $f'(x) = 2x + 0 - 3$ $f'(x) = 2x - 3$
3.4.1	$f(x) = \frac{2}{x}$ $f(x) = \frac{2}{(x+h)}$ $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$ $f'(x) = \lim_{h\to 0} \left[\frac{2}{(x+h)} - \frac{2}{x}\right] \div h$ $f'(x) = \lim_{h\to 0} \left[\frac{2x - 2(x+h)}{x(x+h)}\right] \div h$ $f'(x) = \lim_{h\to 0} \left[\frac{\cancel{-2h}}{x(x+h)}\right] \times \frac{1}{\cancel{h}}$ $f'(x) = \lim_{h\to 0} \left[\frac{-2}{x(x+h)}\right]$

	$f'(x) = \frac{-2}{x(x+0)}$ $f(x) = -\frac{2}{x^2}$
3.4.2	$x > 0$
3.5	$f(x) = -5x^2$ $f(x+h) = -5(x+h)^2$ $f(x+h) = -5x^2 - 10xh - 5h^2$ $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$ $f'(x) = \lim_{h\to 0} \frac{-5x^2 - 10xh - 5h^2 - (-5x^2)}{h}$ $f'(x) = \lim_{h\to 0} \frac{\cancel{-5x^2} - 10xh - 5h^2 \cancel{+ 5x^2}}{h}$ $f'(x) = \lim_{h\to 0} \frac{h(-10x - 5h)}{h}$ $f'(x) = -10x - (0)$ $f'(x) = -10x$

RULES OF DIFFERENTIATION

Activity 4

4.1	$y = \frac{\sqrt{x}}{2} - \frac{1}{6x^3}$ $y = \frac{1}{2}x^{\frac{1}{2}} - \frac{1}{6}x^{-3}$ $\frac{d_y}{d_x} = \left(\frac{1}{2} \times \frac{1}{2}\right)\frac{1}{2}x^{\frac{1}{2}-1} - \left(\frac{1}{6} \times -3\right)x^{-3-1}$ $\frac{d_y}{d_x} = \left(\frac{1}{4}\right)x^{-\frac{1}{2}} - \left(-\frac{1}{2}\right)x^{-4}$ $\frac{d_y}{d_x} = \frac{1}{4x^{\frac{1}{2}}} + \frac{1}{2x^4}$
4.2	$D_x[(x^3 - 3)^2]$ $= D_x(x^6 - 6x^3 + 9)$ $= 6x^5 - 18x^2$
4.3	$y = \frac{4}{\sqrt{x}} - \frac{x^3}{9}$ $y = 4x^{\frac{1}{2}} - \frac{1}{9}x^3$ $\frac{d_y}{d_x} = \left(4 \times -\frac{1}{2}\right)x^{-\frac{1}{2}-1} - \left(\frac{1}{9} \times 3\right)x^{3-1}$ $\frac{d_y}{d_x} = (-2)x^{-\frac{3}{2}} - \left(\frac{1}{3}\right)x^2$ $\frac{d_y}{d_x} = -\frac{2}{x^{\frac{3}{2}}} - \frac{x^2}{3}$
4.4	$D_x[(x - 2)(x + 3)]$ $= D_x(x^2 + x - 6)$ $= 2x + 1$

4.5	$y = x^2 - \frac{1}{2x^3}$ $y = x^2 - \frac{1}{2}x^{-3}$ $\frac{d_y}{d_x} = (1 \times 2)x^{2-1} - \frac{1}{2}(-3)x^{-3-1}$ $\frac{d_y}{d_x} = 2x + \frac{3}{2x^4}$

Activity 5

5.1	$y = (2 - 5x)^2$ $y = 4 - 20x + 25x^2$ $\frac{d_y}{d_x} = 50x - 20$
5.2	$y = \frac{x^6}{4} + 4\sqrt{x}$ $y = \frac{1}{4}x^6 + 4x^{\frac{1}{2}}$ $\frac{d_y}{d_x} = \left(\frac{1}{4} \times 6\right)x^{6-1} + \left(4 \times \frac{1}{2}\right)x^{\frac{1}{2}-1}$ $\frac{d_y}{d_x} = \left(\frac{3}{2}\right)x^5 + (2)x^{-\frac{1}{2}}$ $\frac{d_y}{d_x} = \frac{3x^5}{2} + \frac{2}{x^{\frac{1}{2}}}$
5.3	$D_x\left[4 - \frac{4}{x^3} - \frac{1}{x^4}\right]$ $= D_x(4 - 4x^{-3} - x^{-4})$ $= 12x^{-4} + 4x^{-5}$

	$= \frac{12}{x^4} + \frac{4}{x^5}$
5.4	$y = \left(1 - \sqrt{x}\right)^2$ $y = \left(1 - x^{\frac{1}{2}}\right)^2$ $y = 1 - 2x^{\frac{1}{2}} + x$ $\frac{d_y}{d_x} = -1x^{-\frac{1}{2}} + 1$ $\frac{d_y}{d_x} = -\frac{1}{x^{\frac{1}{2}}} + 1$
5.5	$y = \frac{3}{2x} - \frac{x^2}{2}$ $y = \frac{3}{2}x^{-1} - \frac{1}{2}x^2$ $\frac{d_y}{d_x} = -\frac{3}{2}x^{-1-1} - 1x^{2-1}$ $\frac{d_y}{d_x} = -\frac{3}{2}x^{-2} - 1x^1$ $\frac{d_y}{d_x} = -\frac{3}{2x^2} - x$
5.6	$f(x) = (7x + 1)^2$ $f(x) = 49x^2 + 14x + 1$ $f'(x) = 98x + 14$ $f'(1) = 98(1) + 14$ $f'(x) = 112$

Activity 6

6.1	$D_x[1+6\sqrt{x}]$ $= D_x\left[1+6x^{\frac{1}{2}}\right]$ $= 3x^{-\frac{1}{2}}$ $= \frac{3}{x^{\frac{1}{2}}}$
6.2	$y = \frac{8-3x^6}{8x^5}$ $y = \frac{8}{8x^5} - \frac{3x^6}{8x^5}$ $y = x^{-5} - \frac{3}{8}x$ $\frac{d_y}{d_x} = \frac{1}{x^5} - \frac{3}{8}$
6.3	$y = x^{-4} + 2x^3 - \frac{x}{5}$ $\frac{d_y}{d_x} = -4x^{-5} + 6x^2 - \frac{1}{5}$
6.4	$D_x[a^2x^2 + x]$ $= 2a^2x + 1$
6.5	$f(x) = \frac{x^2-9}{x-3}$ $f(x) = \frac{\cancel{(x-3)}(x+3)}{\cancel{(x-3)}}$ $f(x) = x + 3$ $f'(x) = 1$

6.6	$f(x) = \sqrt{x}\left(x + \frac{1}{x}\right)^2$ $f(x) = x^{\frac{1}{2}}(x + x^{-1})^2$ $f(x) = x^{\frac{1}{2}}(x^2 + 2 + x^{-2})$ $f(x) = x^{\frac{5}{2}} + 2x^{\frac{1}{2}} + x^{-\frac{3}{2}}$ $f'(x) = \frac{5}{2}x^{\frac{3}{2}} + x^{-\frac{1}{2}} - \frac{3}{2}x^{-\frac{5}{2}}$ $f'(x) = \frac{5}{2}x^{\frac{3}{2}} + \frac{1}{x^{\frac{1}{2}}} - \frac{3}{2x^{\frac{5}{2}}}$